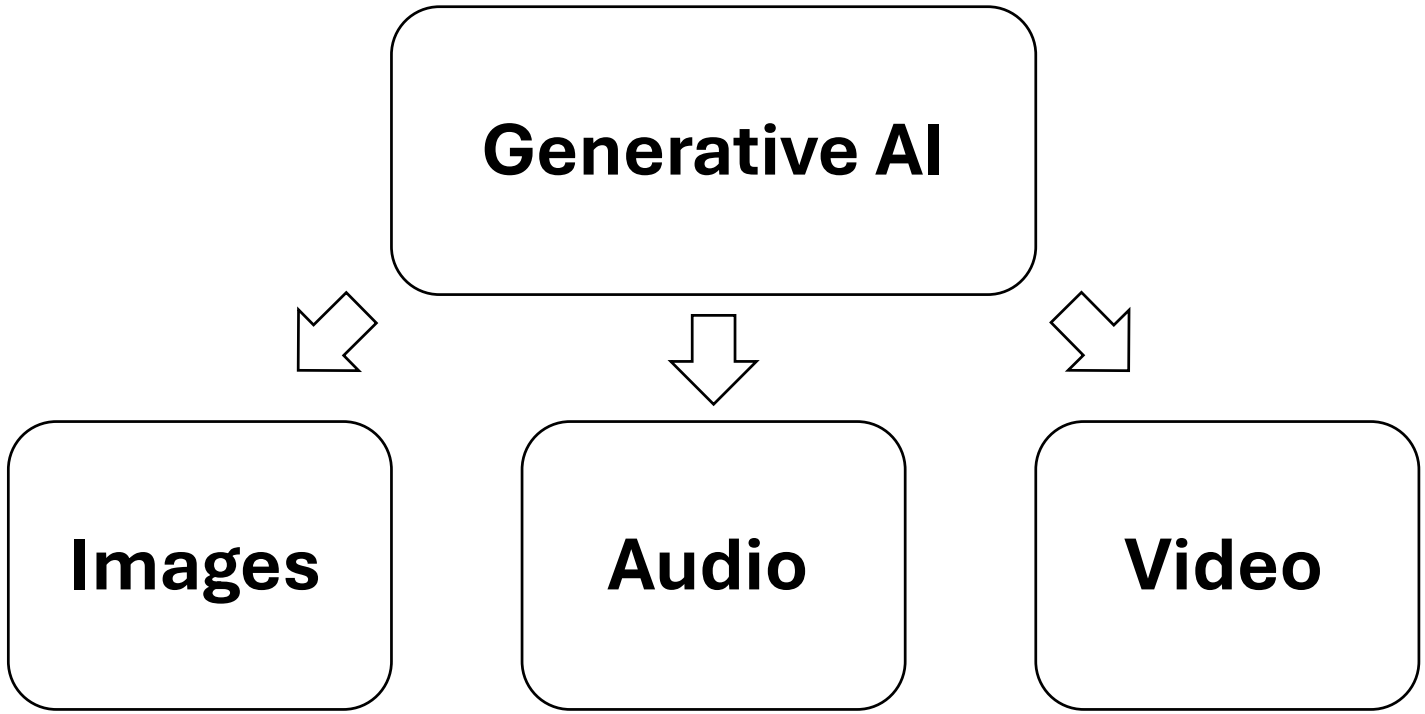
PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

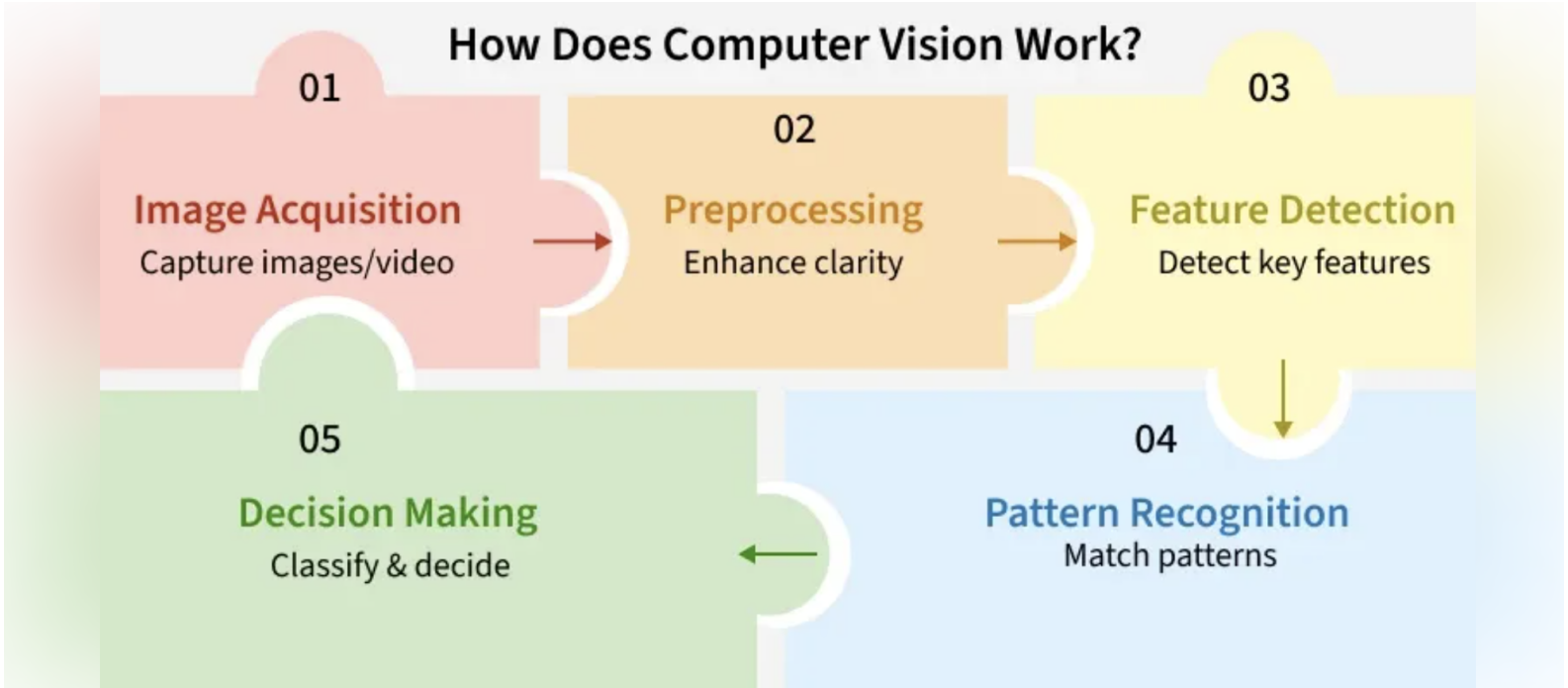
Christopher Zorn

October 19, 2026

Beyond Text



Prequel: Computer Vision



(source)

Computer Vision ↔ Image Generation

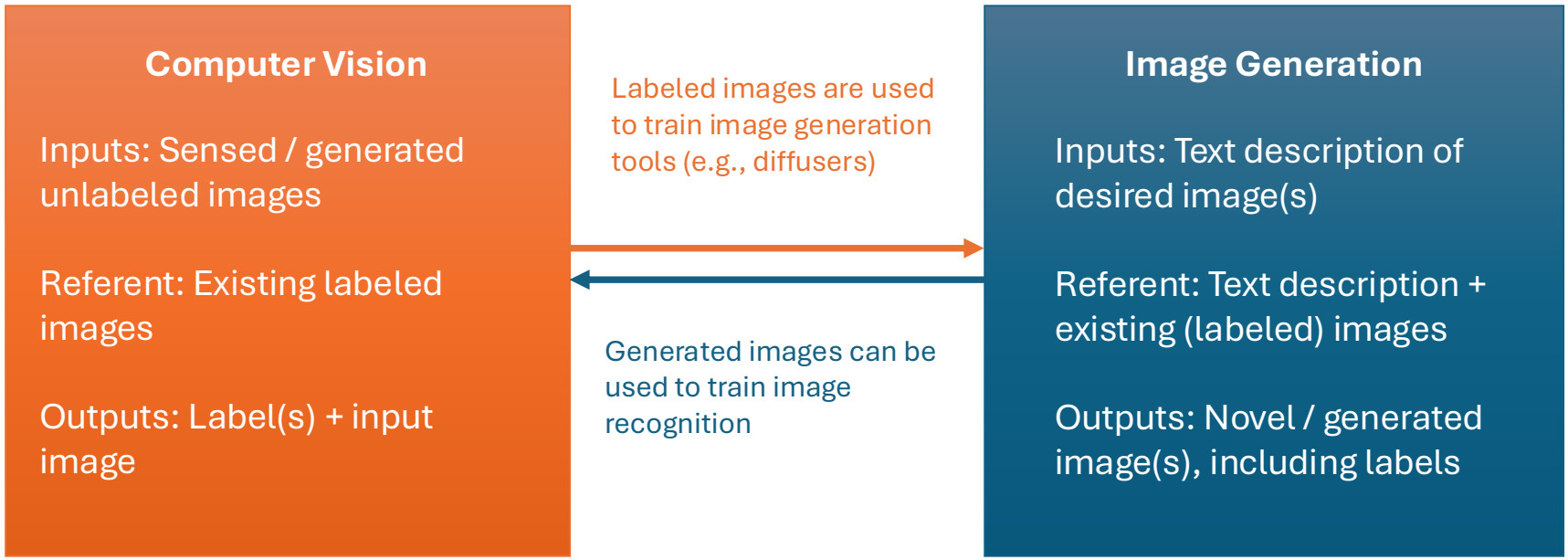


Image Generation: Models

Generative Adversarial Networks (GANs)

- Two-part model:
  - The generator creates images that attempt to be as accurate as possible
  - The discriminator evaluates the images for realism
- Feedback from the discriminator makes the generator's images better
- Improvement in the generator's images helps the discriminator get better at distinguishing realistic from unrealistic images
- Largely superseded by diffusion models (but "adversarial loss" components remain)
- Find it used at (e.g.) https://this-person-does-not-exist.com/

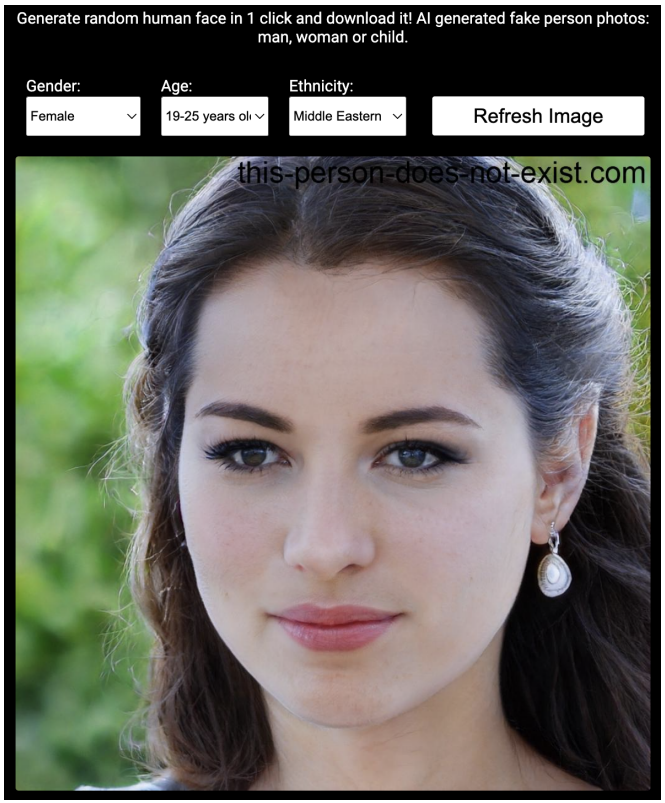
Image Generation: Models

Diffusion Models

- Start with a "noise image," then add detail ("diffuse the entropy away")
  1. Learning: Take an image, add noise, then recreate it
  2. Generation: Start with noise + prompt, add detail (via learning + prompt information)
- Diffusion models are faster, more stable, more versatile than GANs
- Useful explainers: here, here, and here (interactive)
- Examples: Stable Diffusion, Dall-E, Midjourney, etc.

GANs vs. Diffusion Models

GAN Model



Diffusion Model (SeeDream 4.5 via OpenArt.ai)



Prompt: "Create a photorealistic image (head-shot) of a 19-25 year old Middle Eastern Woman."

Generated Music



The process:

1. Train on massive, labeled digital audio data.
2. Training $\rightarrow$ statistical patterns of tonality, instrumentation, etc.
3. Prompt $\rightarrow$ waveform (at requested level of fidelity)

Examples: Suno, Udio, Stable Audio, MusicGen (Meta).

*Spectrogram output of Riffusion (text-to-audio via Stable Diffusion). CC BY-SA 4.0, Wikimedia Commons.

GenAI Music

GPT-4

Can you compose a short tune (say four to eight bars) using ABC notation?
There are many possible ways to compose a short tune in abc notation, but here is one example:

```
X:1
T:Example Tune
M:4/4
L:1/8
K:C
|:G2AB | G2AB c2c2 | edcB c2BA | G4-G2z2 :|
|:c2de f2ed | c2de f2f2 | g2fe d2cB | c4-c2z2 :|
```

This tune has two four-bar phrases, each repeated once, forming an A part and a B part. The key is C major.

Can you describe the tune in musical terms?
Here is one possible description:
The tune starts with a rising arpeggio of the tonic chord (C major), followed by a descending scale that returns to the tonic. This pattern is repeated, but the second time the scale continues to the higher tonic note. The first phrase ends with a long tonic note.
The second phrase begins with a similar pattern, but using the subdominant chord (F major) instead of the tonic. The second time, the scale continues to the higher subdominant note. The phrase ends with a stepwise descent to the tonic, followed by a long tonic note.
The A part and the B part have a similar structure, but use different chords and ranges. The tune has a simple and catchy melody, with a clear sense of tonality and rhythm.

I like part A, but maybe part B is just a bit too similar to part A, making the song repetitive. How about beginning part B with a descending arpeggio instead of a rising one?
Sure, that would create some contrast and variation. Here is one possible way to do that:

```
X:1
T:Example Tune
M:4/4
L:1/8
K:C
|:G2AB c2BA | G2AB c2c2 | edcB c2BA | G4-G2z2 :|
|:f2ed c2BA | f2ed c2c2 | g2fe d2cB | c4-c2z2 :|
```

That's much better. Now how about making this into a duet, can you add a second staff for this with a bass accompanying the melody?
One possible way to write a second staff for this tune is to use root notes and fifths of the chords, as well as some passing notes, to create a simple accompaniment. Here is an example in abc notation, with the second hand bass on the lower staff:

```
X:1
T:Example Tune
M:4/4
L:1/8
K:C
V:1
|:G2AB c2BA | G2AB c2c2 | edcB c2BA | G4-G2z2 :|
|:f2ed c2BA | f2ed c2c2 | g2fe d2cB | c4-c2z2 :|
V:2 clef=bass
|:C,2C,2 G,2G,2 | C,2C,2 G,2G,2 | F,2F,2 E,2E,2 | C,4-C,2z2 :|
|:F,2F,2 C,2C,2 | F,2F,2 C,2C,2 | G,2G,2 F,2F,2 | E,2C,2 G,2C,2 :|
```

The sheet music of the final tune:



GPT-4 composing music in ABC notation from a text description.

CC BY-SA 4.0, Wikimedia Commons.

The Good:

- Convincingly recombines styles, genres, and structures
- "Clone" a specific artist's voice or sound ("Heart on My Sleeve," 2023)
- Constructive uses [e.g. *Now and Then* (2023)]

The Bad:

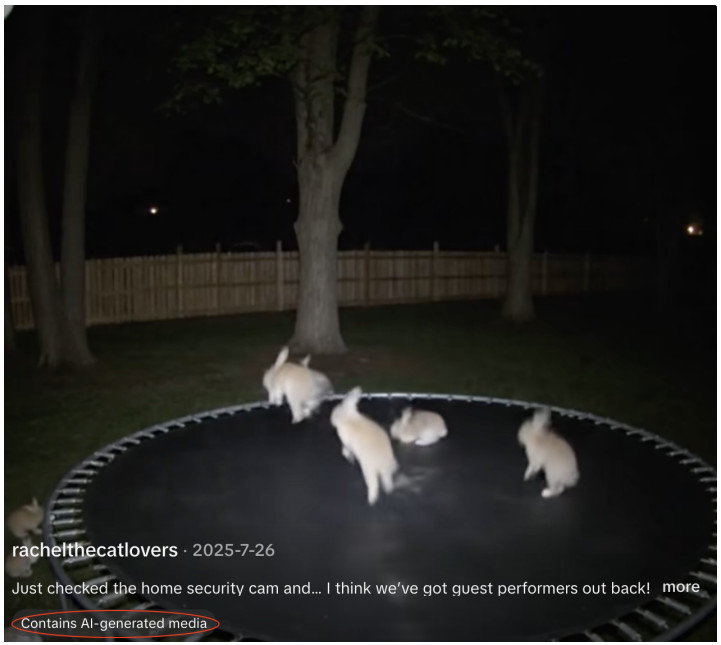
- *Does not* hear, feel, or intend (predicts likely continuations)

$\rightarrow$ Video

Generative video (motion + audio):

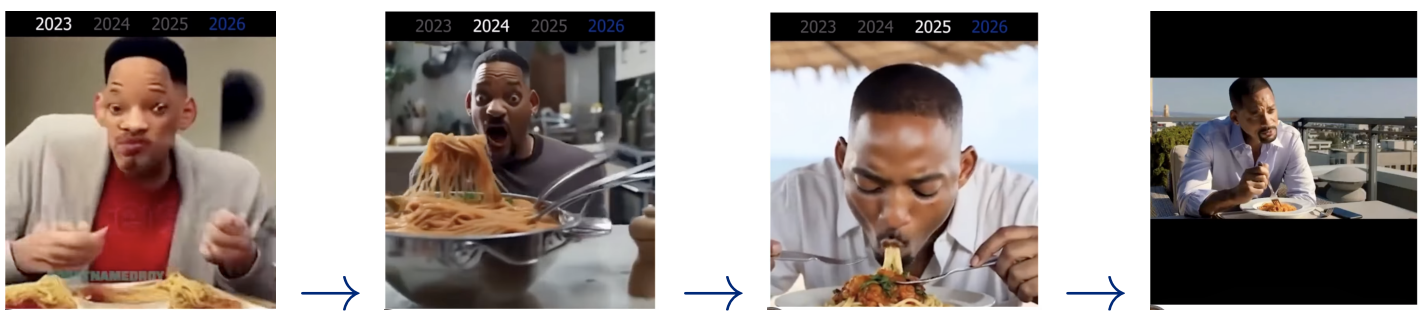
- Same core technology (diffusion + transformers)
- Training data = large video corpora
- Key challenge: *temporal coherence*
  - Transformers applied to a series of images
  - Diffusion models extended over time
- Computational demands are high (relative to text / images / audio)

Video (II)



Approaches:

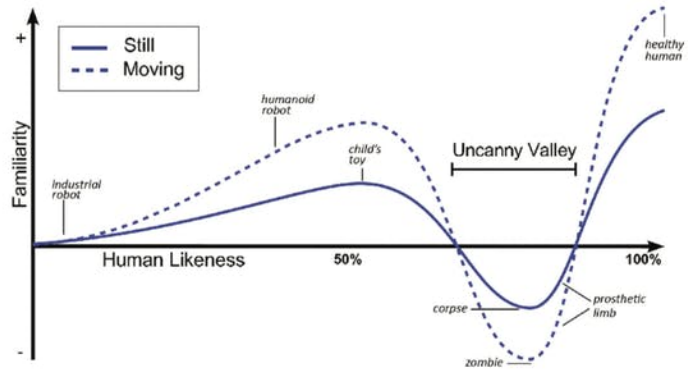
- Text-to-video (Sora, Runway)
- Image-to-video (Kling, Pika)
- Video-to-video (e.g., Luma)
- Inpainting / outpainting (e.g., Veed)
- Avatars (e.g., Synthesia)



The "Uncanny Valley"

From Masahiro Mori (1970):

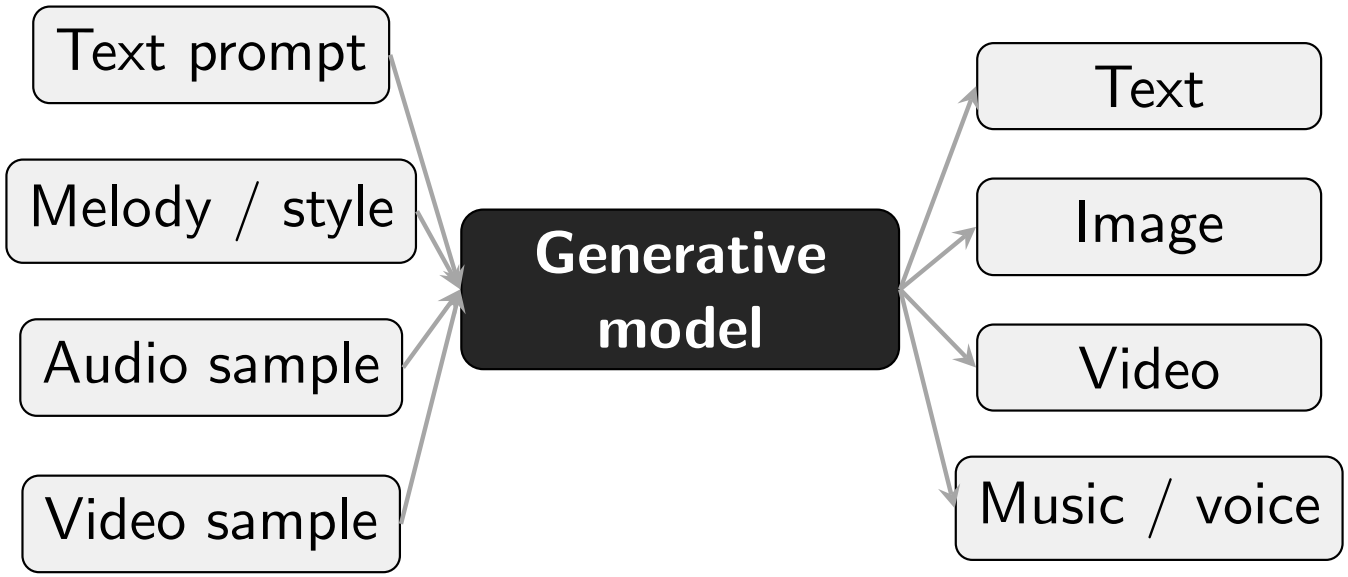
- Near-human likenesses are unsettling
- Motion augments the effect
- Theories: neurological, psychological/cognitive, evolutionary
- Practical impacts (e.g., *The Polar Express* 2004)



(source)



Generative AI: Commonalities



Modality	**Input**	**Example tools**
Text	Text prompt	ChatGPT, Claude, Gemini
Image	Text prompt	DALL·E, Midjourney, Stable Diffusion
Music	Text / style	Suno, Udio, MusicGen
Video	Text / image / video	Sora, Runway, Pika
Voice	Audio sample	ElevenLabs